

Russel's Paradox:-

$$\Rightarrow \underline{A} = \{ X : X \text{ is a set and } X \notin X \} \leftarrow$$

✓ $M = \{1, 2, 3, \dots, 7\}$

$$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

✓ R

~~$B = \{ \}$~~

"Is A an element of A ?"

$x \in A$ it means

~~X~~ ~~Q~~ ~~X~~

$$\rightarrow \frac{A \in A}{\perp}$$

true $A \in A$

False $A \not\subseteq A$

means

it can be

$$A \notin A$$

$A \notin A$

→ resulted in Zermelo Fraenkel axioms

→ well-ordering

→ axiom of foundation

$$X \cap x \neq \emptyset \text{ for all } x \in X$$